

SECTION 01001B-A

QUALITY CONTROL

1.0 QUALITY CONTROL

A. Definitions:

1. Notice of Discrepancy (NOD): Documentation generated by the Engineer, Contractor, or Owner for work installed or in progress which does not conform to Contract Documents.
2. Remedial Action Request (RAR): Documentation generated by the Engineer or Owner for work or materials not in conformance to the Contract Documents and that are in danger of being covered up, , or work that has been identified as deficient in an NOD and not corrected in an expeditious manner.
3. Work List: Contractor-generated list of work that is incomplete, damaged, or identified as unsatisfactory.
4. Punch List: The list of items identified by the Owner and Engineer upon substantial completion that must be completed prior to final completion.
5. Site Observation Report (SOR): A report prepared by the Engineer documenting observations made while visiting the site.

- B. Workmanship: Comply with industry standards, except when more restrictive tolerances or specified requirements indicate more rigid standards or more precise workmanship.

C. Manufacturer's Instructions:

1. Comply with manufacturer's printed and verbal instructions in full detail, including each step in sequence. If instructions conflict with the Contract Documents, request clarification from the Owner before proceeding.
2. Perform work with persons trained and qualified to achieve the workmanship required by the Contract Documents.

D. Factory-Certified Test Logs:

1. Provide factory-certified test logs for products that require such logs that document the name of the quality control inspector, test date, original performance and test data logs, corrective action required, and final certified performance and test data logs.
2. Include test data logs with each shipment for every item that is included in the shipment, enclosed

inside waterproof, clear plastic jackets suitable for placing in a standard three-ring binder.

3. Include with every product item the final certified performance and test data information shown on a permanent nameplate, decal, or sticker located for easy viewing.
- E. Testing Laboratory Services:
1. Provide services of an independent testing laboratory selected by Owner to perform inspections, tests, and other services required by regulatory agencies.
 2. Perform services in accordance with requirements of governing authorities and specified standards.
 3. Submit reports to the Owner and copy the involved subcontractor giving observations and results of tests, indicating compliance or noncompliance with specified standards and the Contract Documents. Initiate corrective action when tests or results indicate noncompliance and retest as appropriate to show compliance as specified.
 4. Cooperate with the testing laboratory personnel and furnish tools, samples of materials, design mix, equipment, storage, and assistance as requested.
 - a. Notify the Engineer and the testing laboratory 24 hours prior to expected time for operations requiring testing services.
 - b. Make arrangements with the testing laboratory and pay for additional samples and tests for the Contractor's convenience.
 5. Uncover work, if so ordered for re-examination by the Owner.
- F. Records: Maintain correct records on an appropriate form for inspections and tests performed, instructions received from the Engineer or testing agency, and actions taken as a result of those instructions. These records shall include evidence that the required inspections or tests have been performed (including type and number of inspections or tests, nature of defects, causes for rejections, etc.), proposed or directed remedial action, and corrective action taken. Document inspections and tests as required by the Specifications.
- G. Quality Control Manual:
1. Furnish, for review by the Owner [at the preconstruction meeting], Contractor's project QC manual proposed to comply with the requirements of the Contract Documents.

2. The project QC manual shall apply to construction operations, both on site and off site, including work by tiered subcontractors, fabricators, suppliers, and vendor representatives, and shall support the proposed construction sequence and schedule. As a minimum, the following shall be provided:
 - a. An organizational chart showing lines of authority for the project QC staff.
 - b. The project QC staff member(s) name, position, and qualifications for that position in resume format. The duties, responsibilities, and authority shall be clearly stated for each QC staff member.
 - c. Acknowledgment of the QC staff responsibility to implement the QC document control systems for all aspects of work specified. A quality program statement of commitment shall be endorsed by Contractor's principle-in-charge and the Owner's project manager.
 - d. Documented authority for the designated QC representative, including authority to momentarily stop work that is not in conformance with the Contract Documents.
 - e. A procedure for tracking, assembling, and storing field QC compliance documentation in a manner which allows functional access for QA auditing by Owner, the Engineer, or independent third-party agencies.
 - f. A procedure for the tracking of construction deficiencies noted in the notice of discrepancy (NOD), remedial action request (RAR), site observation reports (SORs), and third-party inspection reports from identification of discrepancy to an acceptable corrective action implementation.
 - g. A procedure for quarantining damage or nonconforming equipment or materials.
 - h. A list of the definable features of work that identify specific test(s) and the form(s) required for documentation of compliance.
 - i. A procedure for documenting supplier/vendor prequalifications for shelf items.
 - j. A procedure for receiving and inspecting materials and equipment.
 - k. A procedure for daily Contractor's QC observations to include subcontractor work surveillance.
 - l. A procedure for maintaining record drawings.

3. Acceptance of Contractor's project QC manual by the Owner is conditional. The Owner reserves the right to require the Contractor to make changes in the project QC manual, including personnel changes as necessary, to obtain the quality specified in the Contract Documents.
 4. Notify the Owner in writing a minimum of 7 calendar days prior to implementing changes in the approved project QC manual or in the approved Contractor's QC staff. Proposed changes are subject to acceptance by the Owner.
- H. Quality Control Organization: The following list of minimum Contractor's site QC staff qualifications in no way relieves Contractor of meeting the basic requirements of quality construction in accordance with the Contract Documents.
1. Quality Control Organization:
 - a. Identify an individual within the organization as the project QC manager who shall be responsible for the overall management of the project QC manual. The identified individual shall be assigned exclusively to this project and have the authority to act in QC matters for Contractor and minimum qualifications as follows.
 - 1) A college degree in engineering, architecture, construction technology, or construction management with a minimum 5 years of experience on construction projects similar to this Contract, 1 year of the experience to be in a quality control/quality assurance capacity.
 - 2) An individual with 10 years of verifiable construction experience on similar type projects with 1 year of the experience in a quality control/quality assurance capacity.
 - b. The QC manager or approved alternate(s) shall be on site during scheduled construction shifts. Identify alternate(s) to the QC manager in the project QC manual. Periods of absence for the site QC manager shall not exceed 2 weeks at any one time and not more than 20 workdays during a calendar year.
 - c. The QC manager shall report directly to a Contractor executive and shall not answer directly to Contractor's project manager in charge of production or cost on this project. The QC manager shall be considered separate from

and in addition to Contractor's other project management staff. The quality control manager shall not be replaced without the Owner's written consent.

- d. Identify personnel within the organization as alternate QC manager(s). Assign them to the construction site with the authority to act on all QC matters for Contractor in the absence of the QC manager. The identified personnel may perform other duties; however, the principal duties during the QC manager's absence shall be in support of the quality program. The minimum qualifications of alternate QC managers is as follows:
 - 1) A college degree in engineering, architecture, construction technology, or construction management.
 - 2) A minimum 3 years of experience on construction projects similar to this Contract with 1 year of the experience in a quality control/quality assurance capacity.
 - 3) An individual with 8 years of verifiable construction experience in a quality control/quality assurance capacity.
- e. Provide sufficient QC support staff to allow representation on site during scheduled construction shifts. The QC support staff shall be of adequate size to maintain satisfactory coverage of work and have the authority to take appropriate actions required to ensure compliance with the Contract Documents. QC support staff shall have the following minimum qualifications: technically competent and experienced to perform dedicated QC duties and have a high school education with a minimum 1 year of experience in heavy construction.
- f. The principal duty of QC support staff is to control the quality of work; however, these persons may perform other duties.
- g. The QC support staff shall work under the direction of the QC manager for field coverage. The project QC manual will identify the duties and responsibilities of the QC support staff.
- h. Contractor's QC staff members are subject to acceptance by the Owner and the Owner reserves the right to require replacement of members of Contractor's QC staff who in the opinion of the Owner are not performing in the best interest of

the Contractor's accepted QC manual and Contract Documents.

- i. Obtain the Owner's acceptance in writing before replacing members of Contractor's QC staff. Submit qualifications of Contractor's suggested replacement staff to the Owner in resume format.

I. QA/QC Coordination Meeting(s):

1. Prior to starting construction, the Contractor's QC representative shall meet with the Owner to review the Contractor's accepted project QC manual. During the meeting, a mutual understanding of the project QC manual shall be developed, including the documentation to be used for recording the QC operations, control activities, testing, and administration for both onsite and offsite work. Publish meeting notes within 2 days of the meeting. Meeting notes will become a part of the Contract file. Additional QA/QC coordination meetings, if required, may be called by any party to confirm mutual understanding or to address deficiencies in the QC system or procedures which may require corrective action by Contractor.
2. Address quality issues in the weekly contractor meetings conducted by the Owner.

J. Quality Control Procedures:

1. The Contractor is responsible for the quality of the work, the level of quality, and field construction document control. The Contractor's project QC manual is the means by which the Contractor ensures that the construction complies with the Contract Documents including the work and materials of its subcontractors and suppliers. The Contractor's site QC representative shall conduct and direct the quality control efforts for all work.
2. Develop system QC procedures.
3. Implement and monitor quality planning/ coordinating for each definable feature of work.
4. Review the following items, as a minimum, at the preconstruction meeting required under the terms of the Contract or at a specifically scheduled QA/QC kickoff meeting:
 - a. Contract requirements.
 - b. Shop drawings and submittal status.
 - c. Contractor's project QC manual requirements.
 - d. Availability of required materials, equipment, and craft labor.

- e. Contractor's QC inspections, test, observations, and required documentation.
 - f. Familiarity and proficiency of the Contractor and subcontractors' work force to accomplish the required workmanship standards.
 - g. Additional preparatory steps required to accomplish the defined scope of work.
5. Conduct additional preparatory coordination meetings for the same feature of work if the quality of ongoing work is unacceptable to Owner.
 6. Perform daily field observations to ensure continuing compliance to contractual requirements including observation of tiered subcontractors. Record as part of the required QC documentation. Document areas of deficiency and address as required in per paragraph K below.
 - a. Schedule a preacceptance walk-through for the definable feature of work. The Owner shall be given not less than 72 hours' written notification of preacceptance walk-through. Prepare and attach to the notification a work list of incomplete or deficient work. Document additional action items noted during the walk-through and revise the Contractor work list. Distribute the revised work list and track completion of open items.
 - b. Verify that work list(s), SORs, NODs, RARs, and third-party inspection open items are completed prior to scheduling a final acceptance walk-through.
 - c. Facilitate scheduling of startup coordination meetings.
- K. Deficient and Nonconforming Work:
1. The Owner reserves the right to observe ongoing and completed work. This does not relieve the Contractor in any way from the responsibility for quality control.
 2. Notice of Discrepancy:
 - a. Document deficient work by issuing an NOD from the Contractor's QC staff. Number and track notifications. The Owner reserves the right to direct the Contractor to issue an NOD if in its judgment sufficient time has elapsed for compliance to verbal direction without a corrective action resulting.
 - b. Provide a suggested corrective action subject to the Owner's approval prior to implementation. See paragraph 5 below.

- c. Publish weekly an NOD log tracking discrepancies. Transmit the NOD log to the Owner in the Contractor's site QC representative's weekly report. The status of NODs will be a standing agenda item at the weekly Contractor's meeting.
 - d. Document the results upon satisfactory completion of the approved corrective action, and close the NOD as described in paragraph 5 below and so note on the NOD log.
 - 3. The Owner reserves the right to issue a Remedial Action Request (RAR) on deficient work or materials that is in danger of being covered up or deficient work item(s) that have not been corrected in an expeditious manner. Provide a written response to the RAR within 24 hours from initiation.
 - 4. If, after 10 calendar days, the RAR has not generated an acceptable corrective action, the Owner will stop the progress payment for the elements noted as deficient. The stop payment will remain in effect until the RAR deficient item is closed. The status of RARs will be a standing agenda item at the weekly Contractor's meeting.
 - 5. Respond to N.O.D.s and RARs in the form of a proposed corrective action. Implement the approved corrective action upon concurrence from the Owner. Notify the Owner when the item is ready for re-inspection. Reinspect the work and record acceptable results on the NOD or RAR form. Close the NOD or RAR when the correction has been verified and signed off by the Owner. Forward copies of closed NODs and RARs to the Owner in the site QC representative's weekly report.
- L. Quality Control of Product Delivery: Inspections of Products and Equipment.
- a. Inspect products and equipment delivered to the project site for installation by Contractor. Inspect and document items identified by the Owner as requiring observation. Provide adequate notification of pending deliveries of those items.
 - b. Inspect upon receipt, equipment or materials supplied by the Owner for shipping damage, identification, and quality prior to incorporation into the work. Provide written notification of damages to the Owner within 24 hours of receipt for disposition.
 - c. Comply with the Specifications and manufacturers' recommendations for handling and storing of products and equipment. If manufacturer's instructions conflict with Contract Documents,

request clarification from the Owner before proceeding. Document as a discrepancy using an NOD items that have been identified as shipped, handled, or stored improperly and do not incorporate into the work until the manufacturer has inspected the item and validated the corrective action that will enable the warranty to remain in effect.

- d. Quarantine damaged or nonconforming equipment or materials to prevent inadvertent use or installation.

M. Documentation:

- 1. Maintain on the worksite current records for quality control operations, including the work of subcontractors and suppliers. Use approved quality forms. Develop QC documentation forms if required to maintain the level of quality required by the Contract Documents. Observations, inspections, testing, and other QC documentation shall contain factual evidence. Include the following form fields, as a minimum, if project conditions require development of additional QC forms:
 - a. Project name.
 - b. Project location.
 - c. Contractor/subcontractor name.
 - d. Type of QC inspection, test, or observation.
 - e. Date.
 - f. Parameter of test.
 - g. Results and actions required.
 - h. Signature block for Contractor QC staff member observing/participating in the inspection.
- 2. Distribute a weekly site QC report to the Owner signed and dated by the site QC representative. Indicate a description of ongoing work on these reports. Document deficient work with supporting logs/reports. Attach the Contractor field QC observation reports.

- N. Maintain record documents on site. The Owner reserves the right to view the record documents prior to releasing the monthly progress payment. Records not reflecting current field conditions shall be documented as a discrepancy by NOD or RAR.

- O. Submit two copies of field QC documents to the Owner, for project documents, prior to project completion. QC documentation for a definable feature of work shall be

submitted at the same time. The following format shall be used for the submittal:

1. Standard three-ring binder(s), with a slip-in spine label. The spine label shall identify the definable features of the work to which the documents pertain. The binder size and quantity shall be suitable to contain accumulated QC documentation.
 2. Clearly label binder cover with project name and Owner information.
 3. Include a table of contents with chapters divided into the system or definable feature of work.
 4. Include labeled dividers installed between chapters in each binder.
- P. Identify quality forms in the Contractor's project QC manual. Develop the required forms with input from the Owner.

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